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Abstract

Atmospheric aerosols play critical roles in climate and air quality. Accurately assessing their environmental impacts requires understanding aerosol abundance, distribution, and the key factors and processes that control them. Although machine learning has been increasingly adopted in predicting aerosol concentrations, its application in exploring underlying processes is limited, and existing studies often fail to capture processes occurring during preceding days that substantially shape aerosol populations. Here we develop a Lagrangian time-series machine learning framework that better represents those dynamic processes. Three-dimensional back-trajectories of air masses arriving at the receptor site are derived using a physics-driven Lagrangian transport model. Environmental variables along these air mass trajectories are then included as time-series input features for a long short-term memory network to predict aerosol concentrations at the receptor site. We apply this framework to multiyear cloud condensation nuclei (CCN) observations from the eastern North Atlantic (ENA). The resulting model exhibits strong predictive skill and outperforms conventional machine learning models. Moreover, the model trained solely on the ENA data successfully reproduces a large fraction of CCN variability at a site in the South Atlantic, indicating that it indeed captures the general processes driving marine CCN variability. Model interpretation reveals that the seasonal variation of CCN over ENA is primarily controlled by aerosol production associated with short-wave radiation, whereas the variation on a shorter timescale is dominated by wet removal. This Lagrangian time-series framework, which is more physically consistent with actual atmospheric processes, offers a powerful tool for predicting and understanding the mechanisms of other atmospheric components.

A Lagrangian Time-Series Machine Learning Framework for Predicting Concentrations and Exploring Drivers of Atmospheric Aerosols: Model Development and Application to Cloud Condensation Nuclei in Marine Boundary Layer

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Key Points

- A Lagrangian time-series machine learning (ML) framework is proposed to better capture the dynamic processes governing aerosol variability
- It achieves higher predictive skill for cloud condensation nuclei concentrations in marine environments compared to conventional ML models
- Its more physically informed architecture enables identification of the controlling processes and their characteristic timescales

Abstract

Atmospheric aerosols play critical roles in climate and air quality. Accurately assessing their environmental impacts requires understanding aerosol abundance, distribution, and the key factors and processes that control them. Although machine learning has been increasingly adopted in predicting aerosol concentrations, its application in exploring underlying processes is limited, and existing studies often fail to capture processes occurring during preceding days that substantially shape aerosol populations. Here we develop a Lagrangian time-series machine learning framework that better represents those dynamic processes. Three-dimensional back-trajectories of airmasses arriving at the receptor site are derived using a physics-driven Lagrangian transport model. Environmental variables along these airmass trajectories are then included as time-series input features for a long short-term memory network to predict aerosol concentrations at the receptor site. We apply this framework to multiyear cloud condensation nuclei (CCN) observations from the eastern North Atlantic (ENA). The resulting model exhibits strong predictive skill and outperforms conventional machine learning models. Moreover, the model trained solely on the ENA data successfully reproduces a large fraction of CCN variability at a site in the South Atlantic, indicating that it indeed captures the general processes driving marine CCN variability. Model interpretation reveals that the seasonal variation of CCN over ENA is primarily controlled by aerosol production associated with short-wave radiation, whereas the variation on a shorter timescale is dominated by wet removal. This Lagrangian time-series framework, which is more physically consistent with actual atmospheric processes, offers a powerful tool for predicting and understanding the mechanisms of other atmospheric components.

Plain Language Summary

Atmospheric aerosols, i.e., tiny airborne particles suspended in the atmosphere, strongly influence global climate and air quality. These particles are not only shaped by local environmental conditions, but also by what the airmass experienced hours to days earlier as it traveled through the atmosphere. In this study, we develop a machine learning framework that captures the influence of airmass history in a physically realistic way. Instead of relying only on local conditions alone, the model utilizes the time series of influencing factors along the airmass trajectory before arrival as the input. Therefore, the model can learn the key processes acting along the trajectory, improving both aerosol concentration predictions and identification of key controlling processes. We demonstrate this approach using multiyear observations from the eastern North Atlantic. The trained model shows strong predictive skills of cloud-forming particle concentrations and performs substantially better than common machine learning models that use only local conditions or average conditions along the trajectory. Based on the model, we identify the main factors driving aerosol variability on different timescales. Although demonstrated here for aerosols, this framework can also be applied to other atmospheric components that are influenced by airmass history, such as trace gases or cloud properties.

1 Introduction

Aerosol particles are an important component of the Earth system and play critical roles in both global climate and air quality (Seinfeld & Pandis, 2016). Aerosols influence the radiation budget by scattering and absorbing solar radiation (i.e., direct radiative effects) and by modifying the microphysical and macrophysical properties of clouds (i.e., indirect radiative effects) (Ackerman et al., 2004; Albrecht, 1989; Li et al., 2022; Lohmann & Feichter, 2005; Twomey, 1977). The aerosol indirect effects remain poorly constrained and constitute the largest source of uncertainty in radiative forcing over the industrial period (Masson-Delmotte et al., 2021). In addition, elevated aerosol concentrations from anthropogenic emissions, wildfires, and dust storms can lead to severe air pollution in densely populated regions, posing a major global public health risk (Cohen et al., 2017; Li et al., 2023a; Pöschl, 2005). These climate and air-quality impacts are closely linked to aerosol abundance and physiochemical properties. Therefore, a predictive understanding of aerosol properties and their controlling processes is essential for quantifying aerosol radiative effects in past, present, and future climates (Carslaw et al., 2013; Gryspeerdt et al., 2023; Masson-Delmotte et al., 2021), as well as for informing effective air pollution mitigation policies and reducing associated health burdens (Huang et al., 2014; Zhai et al., 2021).

Atmospheric aerosols originate from a wide range of natural and anthropogenic sources that include both primary emissions and secondary formations (Pöschl, 2005; Seinfeld & Pandis, 2016). Their concentrations and distributions in ambient atmosphere are governed by a suite of complex, nonlinear processes, such as particle formation, condensation growth, coagulation, chemical transformation, dry deposition, and wet removal, all of which are modulated by a variety of environmental factors. Despite extensive research, the key environmental factors and processes controlling aerosol variability remain incompletely understood. This knowledge gap contributes to large uncertainties in predicting aerosol concentrations and their impacts on climate through aerosol-cloud interactions, as well as on air quality in diverse environments. Notably, chemical transport models and global climate models often fail to reproduce observed cloud condensation nuclei (CCN) concentrations and variability in remote marine boundary layer (MBL) (Block et al., 2024; Fiddes et al., 2025; Gong et al., 2023; Niu et al., 2025; Wyant et al., 2015), indicating that some key processes are missing or inadequately represented in models.

Previous studies have employed steady-state budget equations to identify the drivers of aerosol variability from field observations (Ghate et al., 2023; Mohrmann et al., 2018; Tie et al., 2015; Wood et al., 2012; Zheng et al., 2018). However, those equations omit many key processes and are often oversimplified, limiting their ability to provide a comprehensive picture. For example, previous studies usually neglected secondary aerosol formation when diagnosing the CCN drivers in remote MBL (Ghate et al., 2023; Mohrmann et al., 2018; Wood et al., 2012), which has been shown to make substantial contributions to CCN formation (Zheng et al., 2020a; Zheng et al., 2018). Moreover, this approach relies heavily on prior knowledge of known mechanisms and relationships, limiting their ability to gain new insights. Statistical methods, such as correlation analysis with relevant environmental factors, have also been employed to infer key drivers (Mansour et al., 2022; Sanchez et al., 2021). However, these methods are less effective at handling multivariate non-linear dynamic systems and commonly lack robust validation.

Machine learning offers a powerful tool for uncovering complex nonlinear relationships in large, multivariate datasets and has been increasingly applied in geoscience research (Chen et al., 2022; Reichstein et al., 2019; Shen et al., 2023). Numerous studies have successfully used machine learning to predict aerosol concentrations (Shen et al., 2024; Wei et al., 2023; Yu et al., 2022; Zhou

et al., 2024b). However, applications focused on elucidating the underlying physical processes remain limited. While some studies have combined machine learning with model interpretation techniques, such as partial dependence plot and Shapley values, to identify key influencing factors (Grange et al., 2018; Hou et al., 2022; Houdou et al., 2024; Li et al., 2025; Song et al., 2022; Stirnberg et al., 2021), their model structures generally do not adequately capture the spatiotemporally dynamic nature of atmospheric aerosol processes. Fine-mode aerosols, which are especially relevant for aerosol-cloud interactions and human health, typically have lifetimes of several days, during which processes such as primary emissions, secondary formation, and removal continuously shape aerosol population in the airmass as it travels thousands of kilometers (Clarke et al., 2013; Wang et al., 2020). Consequently, aerosol concentrations at any given location and time are influenced not only by local conditions but also by the processes in preceding days during the transport. Most previous machine learning studies use instantaneous, local variables alone as input features, thereby cannot adequately capture the processes during preceding days (Li et al., 2023b; Nair & Yu, 2020; Stirnberg et al., 2021; Yao & Zhang, 2024). Although a few studies have incorporated limited trajectory-based information (e.g., airmass cluster type, time fraction over different surface types) (Hou et al., 2022; Li et al., 2025; Song et al., 2022), they still do not capture the full temporal evolution of influencing factors. To improve predictive performance, local aerosol composition measurements (Nair & Yu, 2020) and/or concentrations simulated by chemical transport models (Nair et al., 2021; Yu et al., 2022) are often included as input features. However, such approach limits mechanism understanding and impedes the discovery of new process-level insights, especially those that occur during transport.

To better represent the spatiotemporally dynamic processes, we propose a Lagrangian time-series machine learning framework for aerosol prediction and analysis of controlling processes. Leveraging long short-term memory (LSTM) networks (Hochreiter & Schmidhuber, 1997) – a class of recurrent neural networks well-suited for time-series tasks – our approach incorporate the temporal evolution of influencing factors along airmass trajectory as input features. Instead of relying solely on local conditions, the model explicitly tracks atmospheric histories to predict aerosol concentrations at the receptor site. We have applied this approach and trained a model using long-term CCN observations in the remote Atlantic Ocean, the resulting model demonstrates high predictive skill and strong generalizability. Based on the trained model, we further investigate the dominant drivers of CCN variability and their characteristic timescales. This framework, which integrates physics-driven airmass backward trajectory modeling with data-driven LSTM, more faithfully captures aerosol evolution in the atmosphere and enables deeper process-level understanding. Because all input features are derived from globally available datasets, this framework can be readily extended to other regions. Moreover, it can be applied to improve the predictions and investigate the drivers of other atmospheric constituents, such as gaseous species and cloud properties that are strongly influenced by airmass history.

2 Methods and Data

2.1 A Lagrangian Time-Series Machine Learning Framework

As discussed above, the atmosphere is a highly dynamic system, and the abundance and properties of aerosols at a given location are strongly influenced by a variety of processes in the airmass during preceding days. The core idea of our Lagrangian time-series machine learning framework

is to emulate these atmospheric processes by: (1) tracing the air mass movement before it arrives at the observation site, (2) extracting time series of relevant variables along the air mass trajectory, and (3) using these time series as input to an LSTM model to predict aerosol concentrations at the observation site. A schematic diagram of this framework is depicted in Figure 1.

Here, we assume a total of N samples and m input variables, with each input variable represented by a time series of T timesteps along the air mass trajectory. For each sample, the input multivariate time series $x = (x_1, x_2, \dots, x_T) \in \mathbb{R}^{m \times T}$, in which $x_t \in \mathbb{R}^m$ is a vector of all m input variables at timestep t . The predicted target value $y = f_\theta(x)$, where f_θ represents the LSTM model with trainable parameters θ . The detailed structure and calculation inside an LSTM model can be found in Hochreiter and Schmidhuber (1997). Notably, the LSTM hidden states and memory cell states at a specific timestep depend on the states at previous timestep and the input vector at current timestep. This mechanism closely mirrors aerosol-related processes in the ambient atmosphere. For example, in a given air parcel, the aerosol concentration at a specific moment depends on its concentration at the preceding moment and the changes that occur in between, which is largely driven by meteorological conditions, atmospheric chemistry, and aerosol dynamics at that time.

In this study, we use the Hybrid Single-Particle Lagrangian Integrated Trajectory (HYSPLIT) model (Stein et al., 2015) to calculate the 10-day air mass backward trajectory arriving at the observation site for every hour during the studying period. The trajectory height at the observation site is set to be 200 m, and the calculation is driven by the Global Data Assimilation System (GDAS, $1^\circ \times 1^\circ$) meteorology fields (<https://www.ready.noaa.gov/data/archives/gdas1/>). With a 1-hour resolution between trajectory points, each trajectory comprises 241 timesteps (i.e., $T = 241$). The selected features and procedures for extracting their values along each trajectory are detailed in Section 2.3. The configuration, training, and evaluation of the LSTM model are described in Section 2.4.

While a few prior studies have applied LSTM to air pollution forecasting, they are based on different frameworks and still do not fully capture the atmospheric processes along air mass trajectory. For instance, Wu et al. (2020) used past local time series – including the target variable itself – to predict the $\text{PM}_{2.5}/\text{PM}_{10}$ ratio, which neglects the spatial movement of air masses. Other studies combined convolutional neural network (CNN) and LSTM using past time series from both local and neighbor sites (Qiu et al., 2023; Wen et al., 2019), which may implicitly account for regional transport but still rely on past target values as inputs. This approach restricts physical interpretability and limits application in scenarios when historical measurements of the target variable are unavailable.

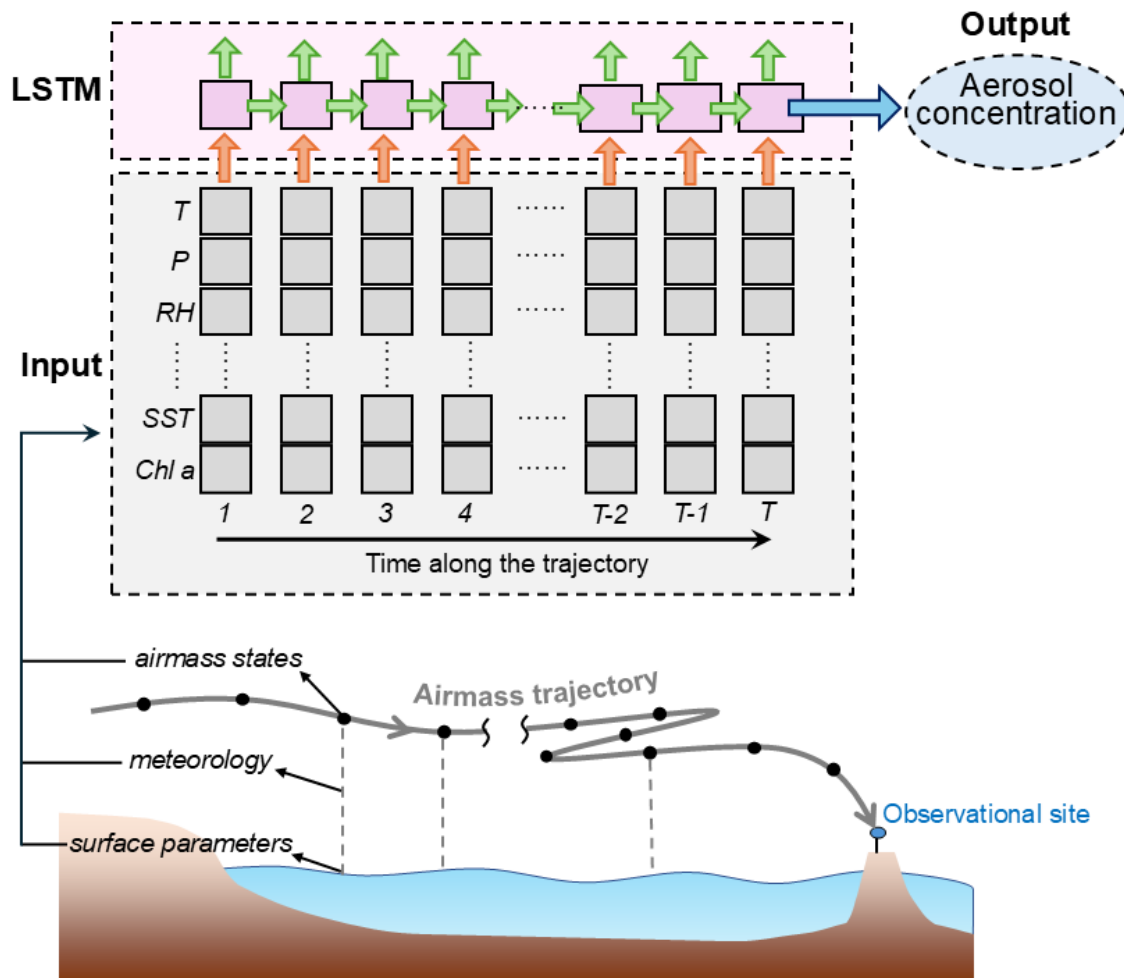


Figure 1. Schematic diagram of the Lagrangian time-series machine learning framework

2.2 Aerosol Measurements

We apply the Lagrangian time-series machine learning framework to train a model that predicts CCN in remote MBL. The CCN data for model training are from the eastern North Atlantic (ENA) atmospheric observatory (39.09° N, 28.03° W, 30 m above mean sea level), located on Graciosa Island in the Azores, Portugal. Situated over 1,500 km from the nearest continent, this site offers an ideal setting to study the CCN budget in the remote MBL (Wang et al., 2022; Wood et al., 2015). Established and operated by the U.S. Department of Energy Atmospheric Radiation Measurement (ARM) program since 2013 (Mather & Voyles, 2013), the site provides long-term aerosol, cloud, and meteorological observations (Uin et al., 2019). CCN concentrations (N_{CCN}) were measured by a cloud condensation nuclei counter (CCNC; Droplet Measurement Technologies), and data from October 2013 to December 2023 are used in this study. Every hour the instrument cycles through 5 supersaturation levels (0.1% – 1.0%), maintaining each level for ~10 minutes. Here, we focus on CCN concentration at 0.2% supersaturation, representative of typical maximum supersaturations in MBL clouds (Gong et al., 2023). The raw 1-second resolution data were averaged over the last 5 minutes of each 10-minute measurement window at 0.2% supersaturation, resulting in an hourly time resolution for training of the machine learning model.

Several procedures are implemented to ensure the quality of the CCN data. These includes (1) removing contamination from local anthropogenic emissions, (2) removing the impact of sea spray aerosols from shoreline wave breaking (Zhou et al., 2025), and (3) checking and correcting for long-term inconsistencies due to instrumental drift and malfunctions. Details of the quality control process are provided in Text S1 in Supporting Information S1.

We also use CCN data from Ascension Island (7.97° S, 14.35° W, 340 m above mean sea level) in the tropical South Atlantic to evaluate the model’s generalizability. These observations were conducted during the Layered Atlantic Smoke Interactions with Clouds (LASIC) campaign (June 2016 – October 2017) using ARM’s Mobility Facility 1 (AMF1) (Zuidema et al., 2016). Similar data preprocessing approaches are performed to obtain the hourly N_{CCN} at 0.2% supersaturation. The final datasets include 40,054 valid hourly CCN observations from ENA and 8,594 from Ascension Island, with the corresponding time series shown in Figure 2.

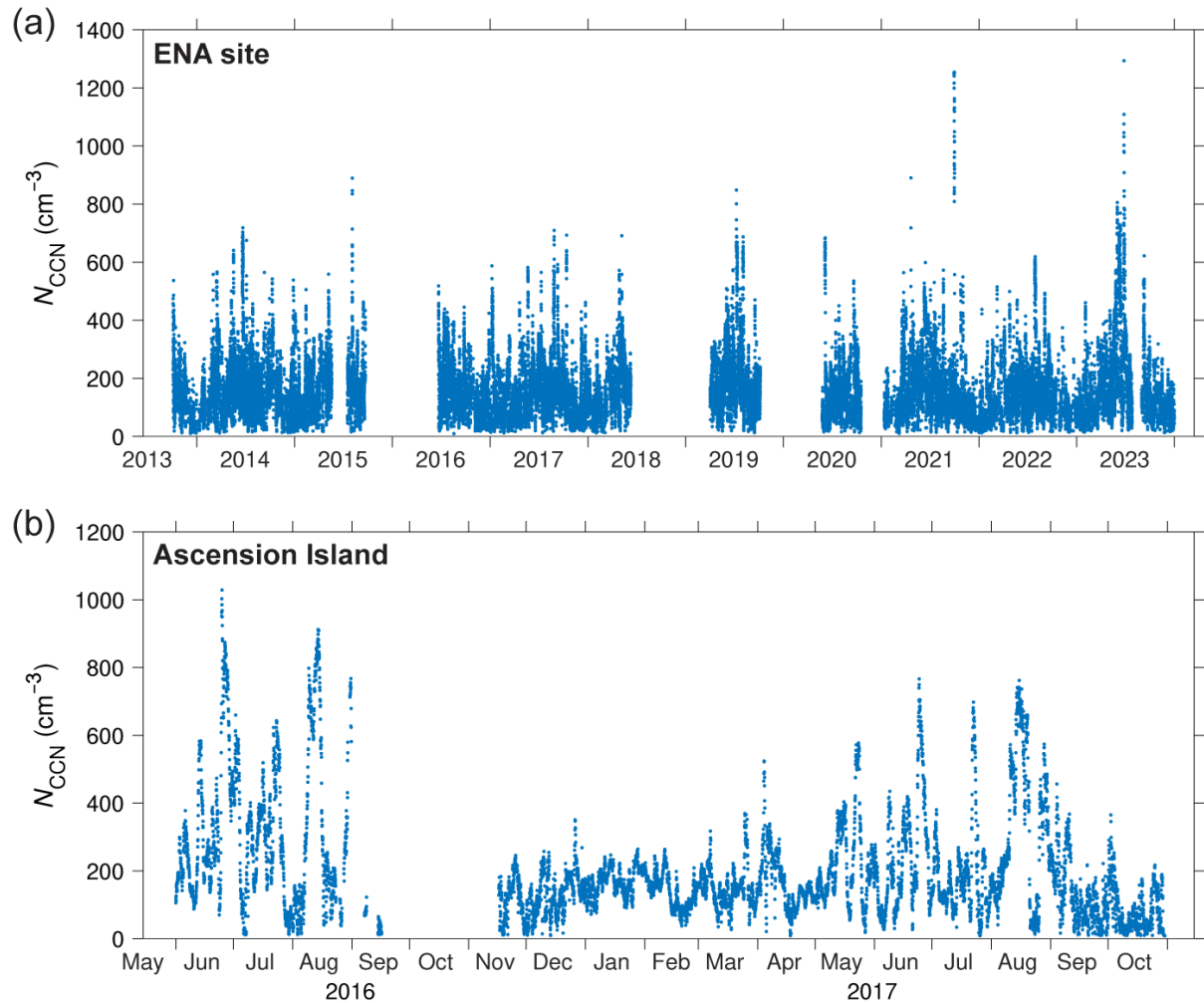


Figure 2. Time series of observed CCN concentrations (N_{CCN}) at 0.2% supersaturation (a) from October 2013 to December 2023 at the ENA site and (b) from Jun 2016 to October 2017 at Ascension Island.

2.3 Input Features and Data Extraction

A total of 22 variables that potentially influence aerosol-related processes and subsequent aerosol concentrations are selected as input features for the Lagrangian-LSTM model. These variables fall into three categories: surface parameters, meteorological conditions, and air mass states (Table 1). The selected surface parameters capture major natural and anthropogenic emission of aerosols and precursors. For example, the SO₂ emission flux (SO₂EM) provides continental and maritime shipping sources of SO₂ and serves as a proxy for other anthropogenic emissions that are highly correlated with SO₂ on global scale. Sea surface concentrations of dimethyl sulfide (DMS) and chlorophyll *a* (CHL), along with surface temperature and wind speed, capture the marine biogenic emissions (McCoy et al., 2015; Zhou et al., 2024a). Surface wind speed is also a key driver of sea spray aerosol production in open oceans (de Leeuw et al., 2011). The meteorological category includes altitude-independent variables characterizing the regional synoptic conditions where the air mass is located, while the air mass state variables capture the physical and chemical properties at the same location and altitude of the air mass. These two categories together allow for capturing the influences of regional synoptic conditions, boundary layer dynamics, free troposphere entrainment, photochemistry, and aerosol wet scavenging, etc.

All input variables, except for surface type flags and air mass pressure, are sourced from globally available gridded datasets, including remote sensing products, reanalysis datasets, emission inventories, and machine learning-derived sea surface DMS fields (Zhou et al., 2024a). Therefore, in-situ measurements are not required to obtain input data for the model. Detailed information of each dataset is provided in Table 1 and Text S2 in Supporting Information S1. We try to exclude the reanalysis concentrations of trace gases and aerosol species as input features because they can be highly uncertain. More importantly, relying on these inputs would make it difficult to investigate the processes that control CCN concentration with the model. However, we make two exceptions: carbon monoxide (CO) and submicron dust (DUST). These species are relatively chemically stable, better simulated in reanalysis, and more interpretable in terms of atmospheric processes. CO serves as a tracer of long-range transport of biomass burning and anthropogenic materials, while DUST indicates the presence and transport of mineral dust aerosols.

To extract time series of surface and meteorological parameters along a given air mass trajectory, we perform a 3-dimensional (latitude, longitude, and time) matchup between trajectory points and the corresponding gridded datasets. For air mass state variables (excluding pressure), the pressure (i.e., altitude) is included as the 4th dimension, with data being interpolated to the pressure level of the air mass. Furthermore, for high-resolution datasets (i.e., SO₂EM, CHL, and precipitation), spatial dispersion of the air mass is accounted for by allowing the spatial search radius to vary with trajectory time (i.e., the time relative to when air mass arrives at the site). Further details on this procedure can be found in Text S3 in Supporting Information S1. As a result, each aerosol concentration measurement is associated with 22 input time series, each with 241 timesteps. Given that many variables exhibit highly skewed distributions, we apply a power transformation to make their distribution more closely resemble a normal distribution or uniform distribution. The exponential values for each variable are listed in Table S2 in Supporting Information S1. The overall data distribution and its evolution with trajectory time of each input feature after the transformation are shown in Figures S9-S10.

Table 1. Information on the input features for the Lagrangian-LSTM model

Category	Variable	Full name	Unit of raw data	Data source	Spatial resolution	Time resolution	Note
Surface parameters	WS	10-m wind speed	m s^{-1}	MERRA-2	$0.625^\circ \times 0.5^\circ$	1 hour	
	TS	Surface temperature	$^\circ\text{C}$	MERRA-2	$0.625^\circ \times 0.5^\circ$	1 hour	
	CHL	Sea surface chlorophyll <i>a</i> concentration	mg m^{-3}	Satellite remote sensing	$0.042^\circ \times 0.042^\circ$	1 day	
	DMS	Sea surface dimethyl sulfide concentration	nmol L^{-1}	ML-derived global daily distribution	$1^\circ \times 1^\circ$	1 day	
	SO2EM	SO ₂ emission flux	$\text{kg m}^{-2} \text{s}^{-1}$	CAMS emission inventory	$0.1^\circ \times 0.1^\circ$	1 day	Combine CAMS-GLOB-SHIP and CAMS-CLOB-ANT considering temporal profiles and COVID-19 adjustment factors
	LAND	Binary flag denoting whether the point is above the land or not	unitless	Classified by the location of trajectory point	-	-	1-over land, 0-over ocean
Meteorology	SLP	Sea-level pressure	hPa	MERRA-2	$0.625^\circ \times 0.5^\circ$	1 hour	
	LTS	Lower tropospheric stability	K	Calculated based on relevant parameters from MERRA-2	$0.625^\circ \times 0.5^\circ$	1 hour	
	EPV	Ertels potential vorticity	$\text{K m}^2 \text{kg}^{-1} \text{s}^{-1}$	MERRA-2	$0.625^\circ \times 0.5^\circ$	1 hour	
	PBLH	Boundary layer height	m	MERRA-2	$0.625^\circ \times 0.5^\circ$	1 hour	
	SWGDN	Surface incoming shortwave flux	W m^{-2}	MERRA-2	$0.625^\circ \times 0.5^\circ$	1 hour	
	CFLOW	Low-level cloud fraction	1	MERRA-2	$0.625^\circ \times 0.5^\circ$	1 hour	
	CFMID	Mid-level cloud fraction	1	MERRA-2	$0.625^\circ \times 0.5^\circ$	1 hour	
	LWP	Liquid water path	kg m^{-2}	MERRA-2	$0.625^\circ \times 0.5^\circ$	1 hour	
	PREC	Precipitation rate	mm hr^{-1}	GPM IMERG V07	$0.1^\circ \times 0.1^\circ$	0.5 hour	
Airmass states	P	Pressure	hPa	HYSPLIT-derived	-	-	The input field for HYSPLIT is GDAS ($1^\circ \times 1^\circ$)
	T	Air temperature	$^\circ\text{C}$	MERRA-2	$0.625^\circ \times 0.5^\circ$	3 hours	Vertical resolution: 72 levels

OMEGA	Vertical pressure velocity	Pa s^{-1}	MERRA-2	$0.625^\circ \times 0.5^\circ$	3 hours	Vertical resolution: 72 levels
RH	Relative humidity	1	MERRA-2	$0.625^\circ \times 0.5^\circ$	3 hours	Vertical resolution: 72 levels
LWC	Liquid water content	kg kg^{-1}	MERRA-2	$0.625^\circ \times 0.5^\circ$	3 hours	Vertical resolution: 72 levels
DUST	Submicron dust mixing ratio	kg kg^{-1}	MERRA-2	$0.625^\circ \times 0.5^\circ$	3 hours	Vertical resolution: 72 levels
CO	CO mixing ratio	mol mol^{-1}	MERRA-2	$0.625^\circ \times 0.5^\circ$	3 hours	Vertical resolution: 72 levels; scaled by observations and EAC4

2.4 Model Configuration, Training, and Evaluation

Our model architecture consists of a single-layer LSTM as the encoder and a fully connected layer as the decoder. Increasing the number of LSTM layers does not significantly improve the model performance. To mitigate overfitting, a dropout layer is incorporated between the encoder and decoder. The model is built based on the Deep Learning Toolbox of MATLAB R2022b.

Model training is based on data collected at the ENA site. Because the aerosol concentration over remote oceans and air mass trajectory path do not change rapidly at hourly timescale, both input and target data from neighbor samples are very similar (i.e., temporal autocorrelation). Therefore, a block splitting strategy is adopted to ensure independence between training and testing data and minimize information leakage (Zhu et al., 2023). Specifically, we set the block size as 1 month, and 6 months in 2022 (Jan, Mar, May, Jul, Sep, and Nov) is used as the testing subset, covering all seasons of a year. For the remaining samples, 15% of days are randomly selected for validation, while the rest are used for training. Here we set block size to 1 day for validation set to balance minimizing data leakage with reaching local minimum in training.

Prior to training, each input feature is standardized using Z-score normalization, and the target variable (i.e., N_{CCN}) is log-transformed to approximate a normal distribution. We utilize Adam as the optimizer with a learning rate of 0.005 and train the model with mini batches. L2 regularization is also adopted to further reduce the risk of overfitting. The mean square error (MSE) is used as the loss function. Training terminates when the maximum number of epochs is reached, and the model parameters corresponding to the lowest validation loss are selected as the final model state. Hyperparameter tuning for the number of hidden units, dropout ratio, L2 regularization factor, and maximum number of training epochs is implemented via grid search. Additional details regarding the tuning process and the selected hyperparameter combination can be found in Text S4 in Supporting Information S1. To enhance prediction robustness, the training-validation split and subsequent training process are repeated for 10 times (Holder et al., 2022; Sigmund et al., 2020), and the final output is obtained by averaging predictions from this 10-member model ensemble.

2.5 Conventional Machine Learning Models for Comparison

To evaluate whether the proposed Lagrangian time-series model significantly improves prediction accuracy, we also train four widely used conventional machine learning models for comparison: random forest (RF), least-square gradient boosting (LSBoost), support vector machine (SVM), and single-layer artificial neural network (ANN). Each model is trained with 14 different sets of input data: (1) the final timestep of the input time series (i.e., local conditions at the ENA site, x_T), using the same data sources as the Lagrangian LSTM model (“Base” in Table 2); (2) the same as the first one, but replacing the data of 6 variables (10-m wind speed, sea level pressure, liquid water path, air temperature, relative humidity, and precipitation rate) with in-situ measurements; (3-8) averages of each input feature over the backward trajectory with durations of 1, 2, 3, 5, 7, and 10 days; and (9-14) exponentially weighted averages over the same trajectory durations, with the weighting factor defined as $w_t = e^{\frac{t-\ell-1}{72}}$, where w_t is the weight for timestep t in a backward trajectory with a length of ℓ hours. For the first two sets of input data, the land flag and air mass pressure are excluded. The data split and model evaluation procedures are consistent with those used for the Lagrangian LSTM model. Hyperparameter tuning is performed individually for each model, with details provided in Text S5 in Supporting Information S1.

2.6 Interpretation Methods

We apply a permutation-based Monte-Carlo global sensitivity analysis to investigate the overall feature importance. Assuming we have two datasets $\mathcal{D}$ and $\mathcal{D}'$:

$$\begin{cases} \mathcal{D} = \{x^{(i)}, y^{(i)}\}_{i=1}^N \\ \mathcal{D}' = \{x'^{(i)}, y'^{(i)}\}_{i=1}^N \end{cases} \quad (1)$$

the input datasets $\{x^{(i)}\}_{i=1}^N$ and $\{x'^{(i)}\}_{i=1}^N$ are from the same population, and each contains N samples and is large enough to represent the characteristics of entire population. y and y' are the corresponding predicted outputs.

For each feature $j \in \{1, 2, 3, \dots, m\}$, a new dataset $\tilde{\mathcal{D}} = \{\tilde{x}^{(i)}, \tilde{y}^{(i)}\}_{i=1}^N$ is constructed based on $\mathcal{D}$ by replacing the data of j -th feature in $x^{(i)}$ with those in $x'^{(i)}$ while keeping all other features unchanged, and the new outputs $\tilde{y}$ are predicted based on $\tilde{x}$.

We then estimate the first-order Sobol' indice (SI_j) as a measure of feature importance (Sobol', 2001; Sobol' & Myshetskaya, 2008):

$$SI_j = \frac{\frac{1}{N} \sum_{i=1}^N (y'^{(i)} - C)(\tilde{y}^{(i)} - y^{(i)})}{D} \quad (2)$$

where C and D are the mean and variance of the concatenated vector of y and y' , respectively.

This analysis does not rely on the availability of true target. In this study, $\mathcal{D}$ is created by using all input data from October 2013 to December 2023 ($N = 89,856$), and $\mathcal{D}'$ is created by randomly permuting the order of samples in $\mathcal{D}$.

This method captures feature importance for overall CCN variability, including both short- and long-term timescales. To assess importance specific to a particular month of year (i.e., no seasonal variability), we repeat the process for each month of year separately. For understanding the dominant drivers for seasonal variability, we treat each month as a unit and apply the similar permutation-based global sensitivity analysis to the monthly-averaged data. Details of the treatment and calculations are provided in Text S6 in Supporting Information S1.

Furthermore, the Lagrangian time-series framework enables tracking how feature importance evolves along the trajectory, offering insight into the temporal dynamics of underlying processes. For each feature $j \in \{1, 2, 3, \dots, m\}$ and each timestep along the trajectory $t \in \{1, 2, 3, \dots, T\}$, we construct a new input dataset based on $\mathcal{D}$, where the values of the j -th feature at timestep t are shuffled across samples while all others remain unchanged. Let y^* denotes the output of the new dataset. The root mean square deviation between y^* and y is used as a metric for the importance of the j -th feature at timestep t ($FI_{j,t}$):

$$FI_{j,t} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y^{*(i)} - y^{(i)})^2} \quad (3)$$

3 Results and Discussions

3.1 Model Performance

3.1.1 Superior Performance of Lagrangian-LSTM over Conventional Machine Learning Models

Conventional machine learning models yield RMSE and R^2 values for testing set $\log_{10}(N_{CCN})$ in the ranges of 0.197 – 0.257 and 0.248 – 0.577, respectively (Table 2). Models trained on local conditions at the ENA site perform the worst, with all testing RMSE values exceeding 0.23 and R^2 values falling below 0.39. These results support our hypothesis that aerosol concentrations are significantly influenced by air mass history and cannot be accurately predicted using local conditions alone. Models incorporating in-situ measurements exhibit modest improvements in performance, which can be attributed to the lower uncertainties of in-situ data compared to reanalysis and satellite-derived inputs.

Substantial performance gains are observed when incorporating information during preceding days along air mass backward trajectories. Model performance initially improves with increasing trajectory length but declines beyond a certain point. The optimal performance is achieved with a trajectory length of 5 days. Furthermore, models that utilize weighted trajectory averages generally outperform those based on unweighted averages. This likely reflects the diminishing influence of earlier air mass history on the final aerosol concentration. Therefore, downweighting older conditions helps improve model accuracy.

Among the four conventional models evaluated, SVM and ANN outperform RF and LSBoost, achieving the best testing RMSE and R^2 of ~ 0.20 and ~ 0.57 , respectively. In comparison, the Lagrangian-LSTM model further improves the performance significantly, achieving a testing RMSE of 0.184 and R^2 of 0.620. These results clearly demonstrate that our Lagrangian times-series framework better captures aerosol-related processes along air mass trajectory and consequently has superior predictive skills relative to conventional static machine learning models.

3.1.2 Weighted Resampling to Mitigate Data Imbalance and the Final Model Performance

Since MSE loss function weights all samples equally, training is dominated by the most common N_{CCN} range, causing the model to underemphasize rarer values. In our dataset, 84% of the N_{CCN} values fall within the range of 50 to 300 cm^{-3} (i.e., $\log_{10}(N_{CCN})$ between 1.7 and 2.5; see Figure S8a in Supporting Information S1). Consequently, training of the model tends to focus on this intermediate N_{CCN} range, resulting in poor prediction performance for extreme N_{CCN} values, which are important for understanding aerosol-related atmospheric processes. As shown in Figure 3a and 3d, the model systematically underestimates N_{CCN} when the true value exceeds 300 cm^{-3} , with a mean $\log_{10}(N_{CCN})$ bias of -0.149 , equivalent to a 29% underestimation in linear space. Conversely, for values below 50 cm^{-3} , the model overestimates with a mean bias of $+0.314$ in log space, corresponding to a 106% overestimation in linear space. As a result, the predicted N_{CCN} distribution is narrower and more concentrated around the intermediate range (Figure 3c).

To mitigate this data imbalance issue due to non-uniform N_{CCN} distribution, we implement a weighted resampling strategy prior to the training-validation split to increase the representation of underrepresented (i.e., extreme) N_{CCN} ranges (Chawla et al., 2002; Haibo et al., 2008; Yu & Zhou, 2021). Details on the resampling methodology are provided in Text S4 in Supporting Information S1. This approach compels the model to better learn from rare events with extreme N_{CCN} values. As a result, the predictive performance for these extreme cases improves significantly (Figure 3b). The mean bias of $\log_{10}(N_{CCN})$ is reduced to -0.116 (a 23.5% underestimation of N_{CCN}) for true $N_{CCN} > 300 \text{ cm}^{-3}$ and to $+0.178$ (a 51% overestimation of N_{CCN}) for true $N_{CCN} < 50 \text{ cm}^{-3}$. Although this resampling slightly degrades overall model performance metrics (RMSE increases from 0.184 to 0.198 and R^2 decreases from 0.620 to 0.604), it yields a more consistent performance across N_{CCN} ranges and a predicted distribution that more closely matches the observations. These improvements are replicated on a different testing subset (i.e., half year of 2021; see Figure S11), suggesting that the benefits of weighted resampling are robust rather than coincidental. Therefore, we adopt the model trained with weighted resampling as our final model for subsequent analyses.

As shown in Figures 4 and S12, the final model effectively captures the temporal variability in N_{CCN} across hourly to seasonal timescales. Most of the pronounced increases and decreases in N_{CCN} are reproduced, albeit with some discrepancies in amplitude and timing. For daily average N_{CCN} , the model explains over 74% of the variance in both logarithmic and linear scales (Figures S13-S14). Overall, given the inherent uncertainties in input datasets and the exclusion of local aerosol concentration measurements as predictors, the model demonstrates satisfactory predictive skill. The consistent model performance across different testing subsets further indicates strong generalizability.

395 **Table 2.** RMSE and R^2 for testing set based on conventional machine learning models with different input data sets and Lagrangian-
396 LSTM model with full time-series as inputs

RMSE	Local conditions		Averages along trajectory						Weighted averages along trajectory						Time series along 10-day trajectory
	Base	With in-situ measurements	1-day	2-day	3-day	5-day	7-day	10-day	1-day	2-day	3-day	5-day	7-day	10-day	
RF	0.257	0.250	0.236	0.229	0.226	0.213	0.224	0.239	0.233	0.229	0.229	0.209	0.208	0.212	0.184
LSBoost	0.245	0.239	0.223	0.217	0.212	0.210	0.219	0.235	0.222	0.218	0.212	0.206	0.207	0.213	
SVM	0.241	0.235	0.220	0.214	0.209	0.204	0.212	0.230	0.221	0.213	0.206	0.197	0.201	0.205	
ANN	0.252	0.243	0.220	0.211	0.203	0.205	0.214	0.228	0.222	0.210	0.202	0.196	0.197	0.203	
Lagrangian-LSTM															
R^2	Local condition		Average along trajectory						Weighted average along trajectory						Time series along 10-day trajectory
	Base	With in-situ measurements	1-day	2-day	3-day	5-day	7-day	10-day	1-day	2-day	3-day	5-day	7-day	10-day	
RF	0.248	0.296	0.365	0.398	0.415	0.490	0.432	0.345	0.383	0.403	0.397	0.505	0.510	0.489	0.620
LSBoost	0.309	0.350	0.432	0.458	0.484	0.496	0.451	0.369	0.436	0.456	0.485	0.514	0.505	0.481	
SVM	0.346	0.384	0.450	0.476	0.505	0.531	0.493	0.400	0.445	0.482	0.518	0.566	0.547	0.523	
ANN	0.284	0.337	0.456	0.506	0.541	0.533	0.494	0.416	0.449	0.512	0.548	0.577	0.568	0.539	
Lagrangian-LSTM															

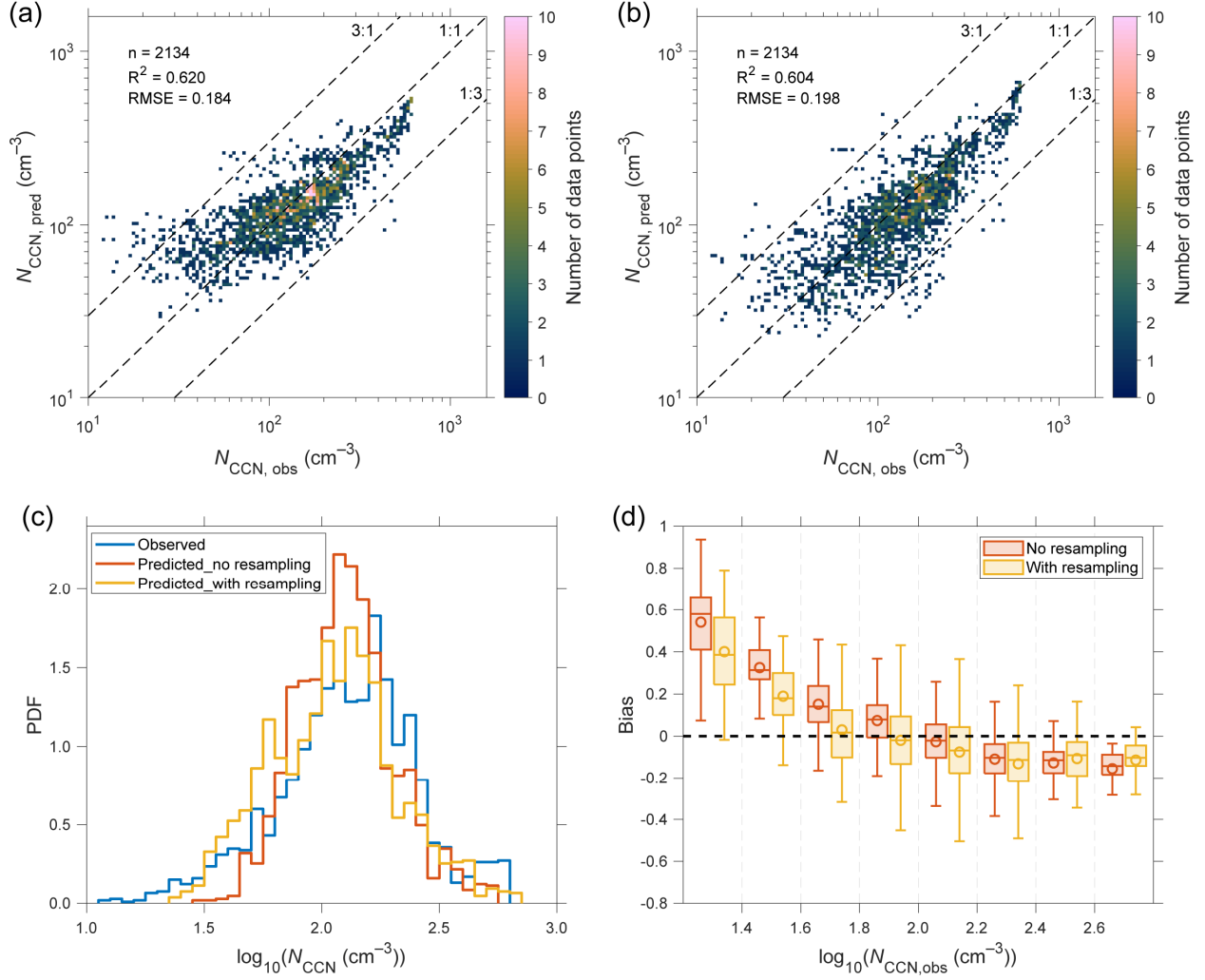


Figure 3. (a-b) Density scatter plots comparing observed N_{CCN} and predictions from the Lagrangian-LSTM models trained (a) without and (b) with the weighted resampling strategy. (c) Probability density functions (PDF) of observed and predicted $\log_{10}(N_{CCN})$, based on models trained without and with resampling strategy. (d) Box plots of the prediction biases (prediction – observation) across different observed $\log_{10}(N_{CCN})$ ranges for both models. In panel (d), boxes indicate the interquartile range (IQR), horizontal lines and circles represent median and mean values, respectively, and whiskers represent the minimum and maximum values within 1.5 times the interquartile range from the lower and upper quartiles. RMSE and R^2 values in panels (a-b) and bias values in panel (d) correspond to $\log_{10}(N_{CCN})$. The data in all panels correspond to the testing set.

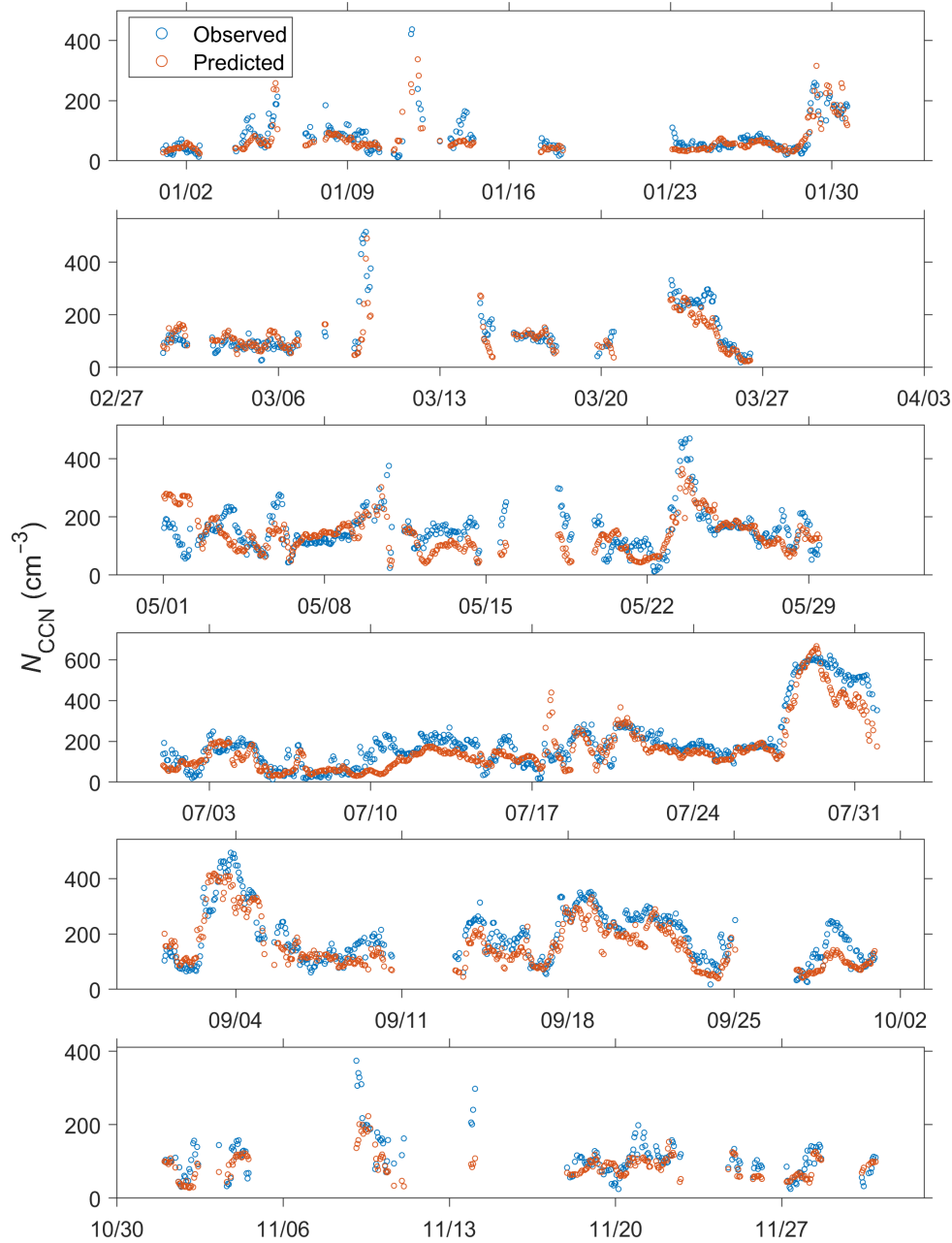


Figure 4. Time series comparison of observed and predicted hourly N_{CCN} for testing set, based on the final Lagrangian-LSTM model developed using the weighted resampling strategy.

3.2 Model Interpretation

3.2.1 Feature Importance and Dominant Drivers for the CCN Variability over ENA

The most influential factors associated with the overall CCN variability across all timescales over the entire measurement period are surface incoming shortwave radiation (SWGDN) and precipitation rate (PREC), followed by surface temperature (TS), liquid water path (LWP), sea-level pressure (SLP), and surface wind speed (WS) (Figure 5a). PREC emerges as the dominant

driver of the intramonth variations of CCN for most months, with the exception of July, during which CO exerts the greatest influence (Figure 5b). LWP also plays an important role in the intramonth variations throughout the year, typically ranking as the second leading factor after PREC. CO and DUST notably influence intramonthly CCN variability in summer and winter, respectively. In comparison, SWGDN and TS have moderate contributions to the intramonthly variability, but they emerge as the top two drivers of longer-term seasonal variations, followed by PREC (Figure 5c).

Given that SWGDN may covary with cloud-related variables (e.g., cloud fraction and LWP) on short timescales at specific locations, its importance based on the LSTM model could stem from correlations with cloud properties. To examine this possibility, we retrain the model after excluding SWGDN from the feature set and observe a statistically significant degradation in model performance (Figure S15). This result confirms that SWGDN provides distinct information that improves the model performance. Furthermore, SWGDN has higher importance for overall CCN variability than any cloud-related variables, and its short- versus long-term importance contrast also differs from those of the cloud-related variables. Together, these results further demonstrate that the significance of SWGDN reflects processes directly related to surface radiation rather than a correlation with cloud properties. A previous machine learning study also identified SWGDN as a key factor governing the variations of aerosol numbers in the Arctic region (Song et al., 2022).

SWGDN primarily influences the production of marine aerosols. For instance, solar irradiance dose has been identified as a key controlling factor for the biological production and spatiotemporal distribution of oceanic DMS (a key precursor of marine sulfate aerosols), particularly in remote low- and mid-latitude oceans (Galí & Simó, 2015; Vallina & Simó, 2007). Similarly, the biological production rate of isoprene, another important precursor of secondary organic aerosols (SOAs), has also been shown to increase with solar irradiance (Gantt et al., 2009; Shaw et al., 2003; Wang et al., 2023). Beyond biological processes, strong radiation can also stimulate abiotic production of isoprene (Ciuraru et al., 2015; Cui et al., 2023) and various other volatile organic compounds (VOCs) at the air-sea interface via photochemical reactions (Bruggemann et al., 2017; Bruggemann et al., 2018; Mungall et al., 2017; Novak & Bertram, 2020). The oxidation of these biogenic or abiotic reactive VOCs and subsequent formation of low-volatility condensable compounds in the atmosphere are enhanced by elevated concentrations of photochemically produced oxidants (e.g., $\text{HO}\cdot$, $\text{BrO}\cdot$) under strong solar radiation (Berresheim, 2002; Seinfeld & Pandis, 2016). These oxidation products can play a pivotal role in new particle formation and subsequent particle growth (Zheng et al., 2020a; Zheng et al., 2021). Consequently, high SWGDN levels promote both the production and oxidation of aerosol precursors and the subsequent formation of secondary aerosols, thereby enhancing the MBL CCN concentrations. This mechanism appears to be a dominant driver of the seasonality of CCN concentration, suggesting the existence of a potential negative feedback loop among surface solar radiation, marine reactive gases, MBL CCN, and marine low-level clouds (Charlson et al., 1987; Vallina et al., 2007).

The high importance of PREC and LWP underscores the critical role of wet removal processes in CCN lifecycle, consistent with prior studies showing that precipitation is a dominant factor driving the variability of N_{CCN} and cloud droplet concentrations in MBL (Mohrmann et al., 2018; Wood et al., 2012). At the ENA site, extreme low-CCN events have been linked to in-cloud coalescence scavenging during cold air outbreaks, where anomalously high LWP values were observed along the air mass trajectories (Wood et al., 2017). Additionally, aerosol budget equation calculations based on ENA measurements also revealed that coalescence scavenging is the dominant sink of CCN or accumulation-mode aerosols (Ghate et al., 2023; Zheng et al., 2018).

The importance of CO during summer reflects the contribution of long-range transported biomass burning and anthropogenic aerosols. This long-range transport has been identified as a sporadic but strong perturbation to aerosol levels at the ENA site and are consistent with the summertime peaks of wildfire activities in North America (Gallo et al., 2023; Wang et al., 2025; Wang et al., 2021; Zheng et al., 2020b). Similarly, the enhanced importance of dust during winter is linked to Saharan dust intrusions, which are often associated with cyclonic dust storms (Gallo et al., 2023; Logan et al., 2014). Additionally, the concurrent transport of anthropogenic pollutants from North Africa and Europe and their interactions with dust particles (i.e., chemical ageing) can further increase N_{CCN} (Astitha et al., 2010; Bègue et al., 2015; Gaston, 2020). Results from the drop-one-feature experiment reveal that removing either CO or dust leads to a significant decline in model performance (Figure S15), indicating their importance is not due to the correlations with other variables. While previous observational studies at this site have identified the influence of these continental aerosol sources in a few isolated cases, our study demonstrates that they exert significant impacts on CCN variability in particular seasons based on multi-year data. The substantially lower feature importance of WS than SWGDN and PREC suggests that the production of sea spray aerosols (SSA) plays a secondary role in the CCN variability, consistent with some of the earlier studies showing SSA has a minor contribution to CCN in remote MBL (Quinn et al., 2017).

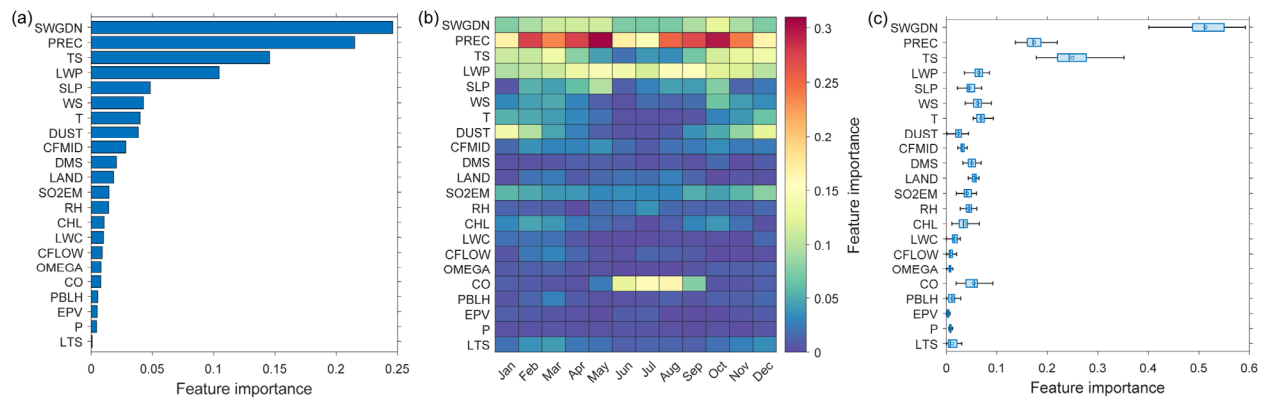


Figure 5. Feature importances (first-order Sobol' indices) for (a) the overall CCN variability, (b) the CCN variability within each month, and (c) the CCN seasonal variability. In panel (c), the boxes represent upper and lower quartiles of the 20 results, the horizontal lines and circles

represent the median and mean values, and the whiskers represent the minimum and maximum values within 1.5 times the interquartile range from the lower and upper quartiles.

3.2.2 Operational Timescales of the Dominant Drivers

As discussed in Section 3.1, the influence of a specific feature on CCN predictions varies with the time elapsed before air mass arrives at the observation site. Leveraging the time-series nature of our Lagrangian-LSTM model, we examine how feature importance evolves along the air mass trajectory and, in doing so, explore the operational timescales of the underlying controlling processes. This analysis not only enhances the interpretability of the model but also contributes to a more mechanistic understanding. Here we focus on the top two drivers of CCN variability (i.e., SWGDN and PREC) to illustrate this approach.

For PREC, the input data corresponding to the final two timesteps along the trajectory exert the greatest influence on the predicted N_{CCN} , and the relative importance of earlier timesteps declines rapidly (Figure 6a). For instance, the importance of PREC one day prior to arrival is only ~25% of that at the final two timesteps. This result is consistent with the more localized effect of precipitation, which can rapidly scavenge activated particles and efficiently reduce MBL CCN population.

In contrast, the peak influence of SWGDN occurs ~10 hours prior to arrival, and its relative importance remains above 0.4 even two days before arrival (Figure 6a). Given SWGDN's strong diurnal cycle and its value being zero at night, we further perform this analysis for each hour of the local solar time at the trajectory point (ToD) to rule out diurnal confounding. The results consistently show the conclusion that SWGDN from ~10 hours prior to arrival is more important than local SWGDN for CCN prediction, regardless of ToD (Figure 6b). Among all combinations of different time-series timesteps and ToDs, the highest feature importance occurs when trajectory time is ~ -10 hours (i.e., ~10 hours before arrival) and ToD is at midday (10:00 – 13:00) when SWGDN is typically the strongest (Figure 6b).

This time-lagged influence of SWGDN is consistent with the role of solar radiation in driving the production of marine reactive VOCs and their oxidations in the atmosphere. Photochemistry-related CCN production involves multiple steps, including VOC emissions, production of low-volatility compounds through oxidation, new particle formation, and subsequent growth to CCN sizes, and therefore it takes some time. The reported global lifetime of atmospheric DMS with respect to chemical reactions ranges from 0.8 to 2.1 days (Chen et al., 2018; Fung et al., 2022; Wohl et al., 2024), though regional variability can be significant depending on oxidant levels and meteorological conditions. The lifetimes of isoprene and monoterpenes are typically shorter, ranging from 0.5 to 5 hours (Kilgour et al., 2024; Kim et al., 2017; Palmer & Shaw, 2005; Shaw et al., 2010). Oxygenated VOCs produced through photochemical reactions at the air-sea interfaces can persist on timescales of several days (Mungall et al., 2017), and a chamber experiment shows that a 12-hour irradiation is required for these VOCs to be further oxidized and promote aerosol

growth from below 20 nm to ~50 nm (Bruggemann et al., 2017). Although our current analysis cannot definitively pinpoint the chemical species or mechanisms driving the time lag in SWGDN's importance, its consistency with established timescales for chemical transformation and particle growth supports this interpretation.

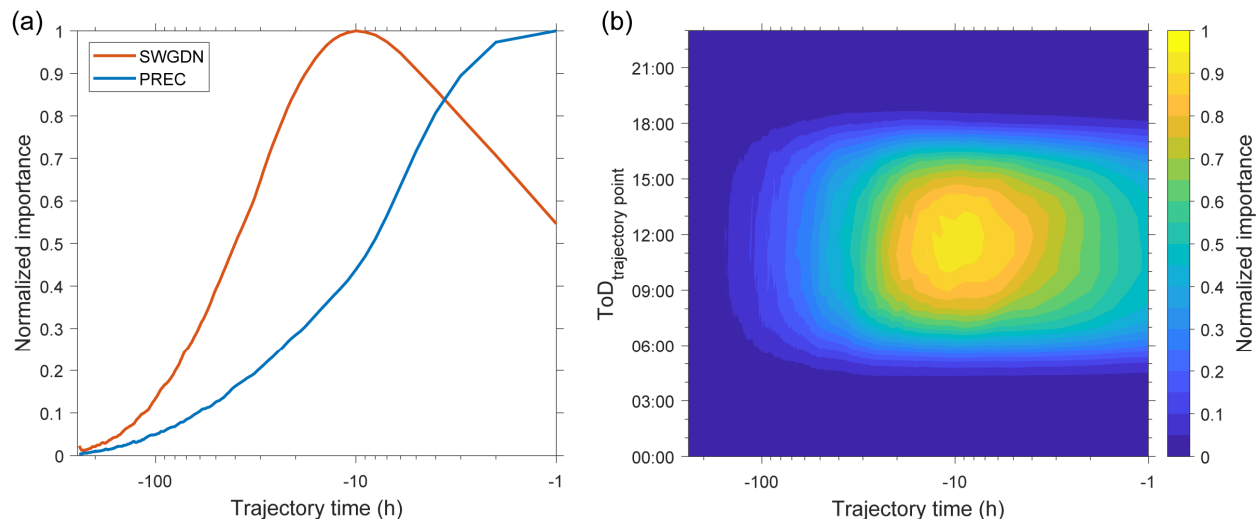


Figure 6. (a) Normalized importance of SWGDN and PREC at each timestep along the trajectory. Trajectory time is defined as the time relative to when airmass arrives at the site. (b) Dependence of normalized importance of SWGDN on the trajectory time and the local solar time at the trajectory point ($ToD_{\text{trajectory point}}$).

3.3 Generalization of the Model to Ascension Island

Beyond demonstrating strong predictive skills on unseen, independent periods at the same location, the LSTM model, trained exclusively on data from the ENA site, also successfully captures a substantial portion of the CCN variability at Ascension Island (Figure 7). Although the predicted absolute values exhibit substantial bias compared to observations throughout the entire LASIC campaign, the model explains ~50% of the temporal variability at an hourly resolution, effectively reproducing most of the pronounced peaks and troughs (Figure 7a,c). The explained variance exceeds 50% when evaluated on the daily averages (Figure 7b,d). This result is particularly noteworthy given that Ascension Island (in tropical South Atlantic) is situated over 5,000 km away from the ENA site (in mid-latitude North Atlantic) and experiences markedly different environmental conditions.

These findings provide strong evidence that the model based on the Lagrangian time-series framework has learnt key mechanisms that control CCN variability in remote MBL. While the testing results in Section 3.1 confirm the model's ability to generalize temporally within the same site without overfitting, they do not necessarily confirm that the model has captured the underlying physical processes. The predictive ability at Ascension Island shows that the model generalizes not only across data space but also in underlying mechanism space, underscoring the robustness of the results from model interpretation.

The observed biases in absolute N_{CCN} values are expected, given the measurement uncertainties, differences in regional background concentrations, and potential variation in the sensitivity of N_{CCN} to specific processes across different environments. As such, while the model effectively captures relative variability, its direct application to predict absolute CCN concentrations across large geographic regions remains challenging and warrants further investigation or site-specific calibration.

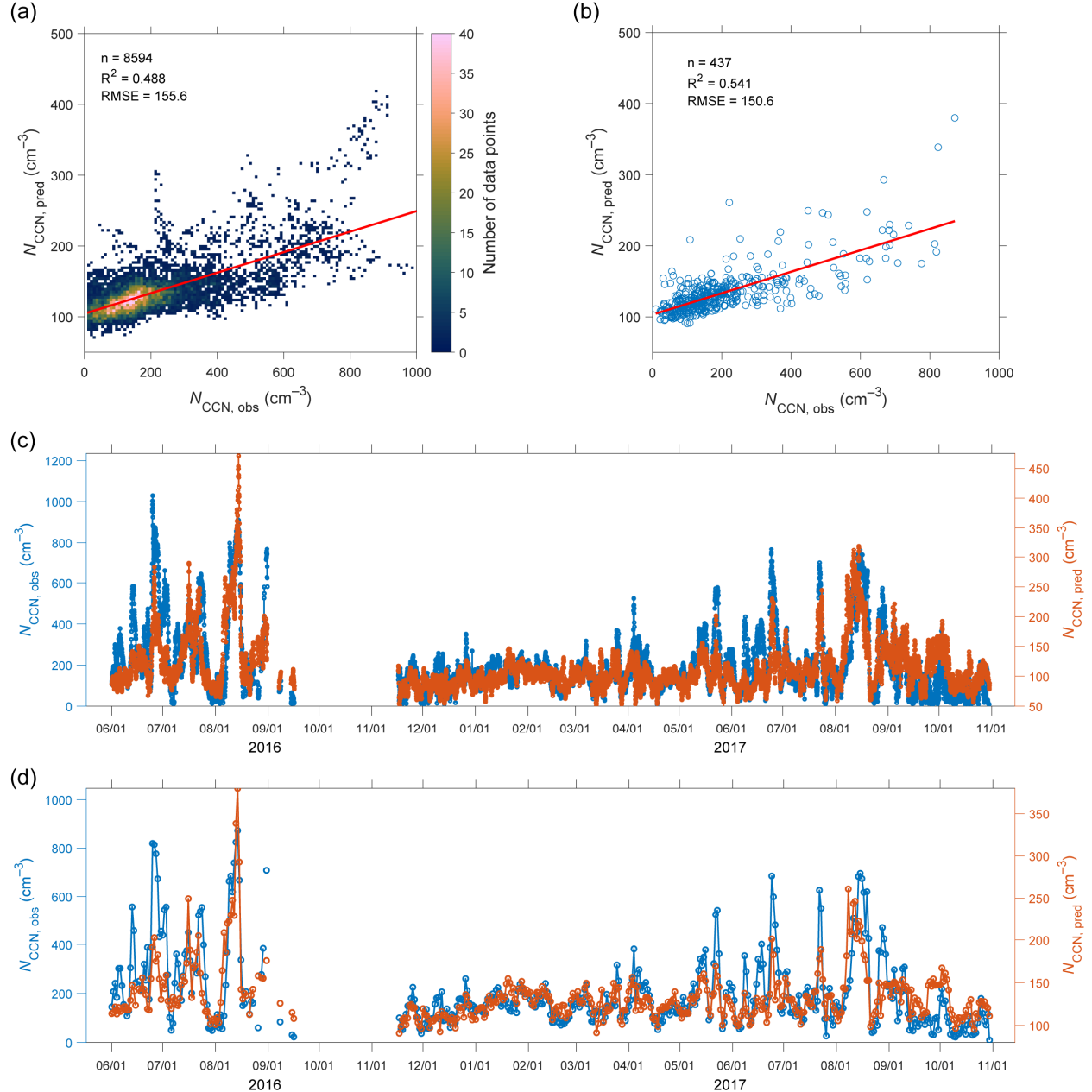


Figure 7. (a) Correlations between observed hourly N_{CCN} at the Ascension Island and corresponding predictions from the model trained on ENA data. (b) Same as panel (a) but for daily averages. Results are shown on a linear scale, as the focus is on the consistency in overall temporal variability rather than relative deviation ratios at individual data points. (c) Time series of observed

and predicted hourly N_{CCN} at the Ascension Island. (d) Same as panel (c) but for daily averages. Note that the Y-axis scales are different between observations and predictions.

3.4 Advantages, Uncertainties, and Limitations

In summary, the Lagrangian time-series framework offers several distinct advantages over conventional machine learning approaches. First, from a process-level perspective, our framework is more physically consistent with actual processes governing aerosol evolution in the atmosphere. While conventional models can use hand-designed features to incorporate earlier information along trajectory (e.g., trajectory averages as shown in this study), such features are often subjective and do not fully capture the spatiotemporal variation of aerosol processes along air mass trajectory. In comparison, our framework exploits spatiotemporal dependencies in a dynamic system more exhaustively (Reichstein et al., 2019), enabling more robust investigations into the controlling processes and their operating timescales. Second, benefiting from the more comprehensive input information and physics-sound structure, the model exhibits a stronger prediction capability and is less susceptible to overfitting. The LSTM encoder extracts key information from a 2-dimensional input matrix, compressing it into a 1-dimensional latent representation, which intrinsically serves as a kind of dimensionality reduction and enhances model generalizability compared to simply flattening the 2-dimensional input matrix and using each element as an individual input feature. Third, this framework represents a paradigm for integrating physics-driven and data-driven approaches. Here we utilize HYSPLIT model to generate air mass backward trajectories, which resolves the spatial dynamics based on fundamental atmospheric transport equations. While it is also possible to represent the spatiotemporally dynamic system with purely data-driven approaches, such as a combination of convolutional and time-series networks (Qiu et al., 2023; Reichstein et al., 2019), they are more computationally intensive and more difficult to interpret. Fourth, all input data originate from globally available gridded datasets, without relying on in-situ measurements. Therefore, this framework is highly transferable and readily applicable to other regions. Moreover, the model structure is not limited to predicting aerosol concentrations. It can be extended to investigate other atmospheric quantities such as aerosol size distributions, trace gas concentrations, and cloud properties, particularly in contexts where recent air mass history is influential.

Despite these advantages, several uncertainties and limitations remain and warrant improvements in future studies. First, currently unavoidable uncertainties exist in input data, impose an upper limit on model performance. These uncertainties stem from both air mass trajectory calculations and source datasets. Trajectory uncertainty generally increases with time before arrival. The uncertainties in gridded source datasets appear to have considerable impact on prediction, as indicated by the significantly better model performance if incorporating in-situ measurements into conventional models. Improving the accuracy and resolution of these source datasets is expected to yield notable improvements in predictive skill. Second, the current set of input features may not capture all relevant processes influencing aerosol variability. This limitation may be due to either

incomplete scientific understanding or the unavailability of suitable datasets. Future studies can systematically test additional features to improve model representation of aerosol processes. Third, the model performance and generalizability could be further improved by increasing the training data volume. While the number of labeled samples (i.e., with valid target value) is constrained by measurement availability, the pool of unlabeled input data can be much larger because all input data are independent of in-situ observations. Incorporating an autoencoder architecture to get a pretrained LSTM using all labeled and unlabeled input data could enhance its ability to learn latent representations, thereby improving downstream performance (Liu et al., 2022; Sagheer & Kotb, 2019).

In terms of model interpretation, the current analysis does not fully disentangle synergistic or correlated effects among input variables. Because mutual correlations can obscure the attribution of importance, it is possible that the importance of a factor is misassigned to another correlated factor. This might be one of the reasons why SWGDN exhibits strong importance but DMS does not. Another potential reason is the large uncertainty in sea surface DMS dataset. To better resolve these relationships, future work should incorporate advanced causal inference techniques capable of identifying and quantifying causal drivers in high-dimensional, time-dependent systems (Runge et al., 2019a; Runge et al., 2023; Runge et al., 2019b). Our Lagrangian time-series framework is structurally well-suited for such causal analysis. In addition to identifying key drivers, it will be useful to further assess the direction and magnitude of aerosol response when a factor is changed or a process is perturbed, which can help quantify the impact of external forcings (e.g., human activity, biomass burning) on MBL CCN population.

4 Conclusions

We propose a Lagrangian time-series machine learning framework to predict the concentration of atmospheric aerosols and exploring the associated drivers. Given the atmosphere is a spatiotemporally dynamic system and fine-mode aerosols exhibit a relatively long lifetime, the aerosol concentration observed at a site is strongly influenced by the history of air mass prior to its arrival. Our framework integrates the physics-driven air mass trajectory model with data-driven time-series machine learning model to emulate these atmospheric processes. We obtain air mass trajectories from HYSPLIT model, extract the multivariate time series of influencing factors along the trajectories, and use these as inputs to an LSTM model to predict the final aerosol concentrations. Trained on multiyear CCN observations at the ENA site, the model demonstrates strong predictive skill during unseen testing periods, outperforming conventional machine learning models trained with either local conditions or trajectory-averaged features. Remarkably, the model also reproduces ~50% of the CCN variability observed at Ascension Island, which is over 5,000 km from the ENA site. This strong transferability suggests that the model captures key universal processes governing aerosol dynamics over oceanic regions.

Model interpretation analyses reveal key drivers of MBL CCN variability over eastern North Atlantic. Seasonal variability is primarily controlled by aerosol production processes associated

with surface short-wave radiation, while the short-term intramonth variations are largely controlled by wet removal, with episodic but strong influences from long-range transported biomass burning aerosols, anthropogenic pollutants, and mineral dust, depending on the season. The Lagrangian time-series framework allows for further investigating the evolution of feature importance along the air mass trajectory. Results show that precipitation within the last 2 hours has the highest importance, consistent with the efficient aerosol removal by precipitation. In contrast, SWGDN ~10 hours earlier exerts the strongest influence of the observed N_{CCN} , likely reflecting its role in modulating the emission and oxidation of marine reactive VOCs, which subsequently affect particle formation and growth to CCN size that take some time.

Overall, the proposed Lagrangian time-series framework provides a more physically sound approach for capturing the complex, history-dependent evolution of aerosol properties compared to static, conventional models. Although our current model does not yet fully disentangle all governing factors and underlying mechanisms, it provides a robust foundation for advancing our understanding of key aerosol-related processes including aerosol-cloud interactions. Furthermore, because the input features can be derived solely from globally available gridded datasets and do not require in-situ measurements, the framework is readily transferable to other regions or extended to study other atmospheric constituents beyond aerosol concentrations.

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Data and Code Availability Statement

All aerosol measurement data and in situ meteorological data used in this study are available in ARM data Archive (<https://adc.arm.gov/discovery/#/>). The MERRA-2 reanalysis datasets and GPM precipitation dataset are available from the NASA Goddard Earth Sciences Data and Information Services Center (<https://disc.gsfc.nasa.gov/>). The satellite-derived sea surface chlorophyll a concentration datasets are available from the Copernicus Marine Service (<https://data.marine.copernicus.eu/products>). All datasets used to construct global daily anthropogenic SO_2 emission fluxes are available from the Emissions of atmospheric Compounds and Compilation of Ancillary Data (ECCAD) website (<https://eccad.sedoo.fr/#/catalogue>). The EAC4 reanalysis dataset is available from the Copernicus Atmosphere Data Store

685 (<https://ads.atmosphere.copernicus.eu/datasets>). The HYSPLIT model and the GDAS
686 meteorological dataset used to drive the model are both publicly available from NOAA Air
687 Resources Laboratory (https://www.ready.noaa.gov/HYSPLIT_hytrial.php;
688 <https://www.ready.noaa.gov/data/archives/gdas1/>). MATLAB scripts for data processing,
689 machine learning model development, and model interpretation have been deposited to Zenodo
690 (<https://doi.org/10.5281/zenodo.18120652>).

691

692 **Conflict of Interest**

693 The authors declare there are no conflicts of interest for this manuscript.

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